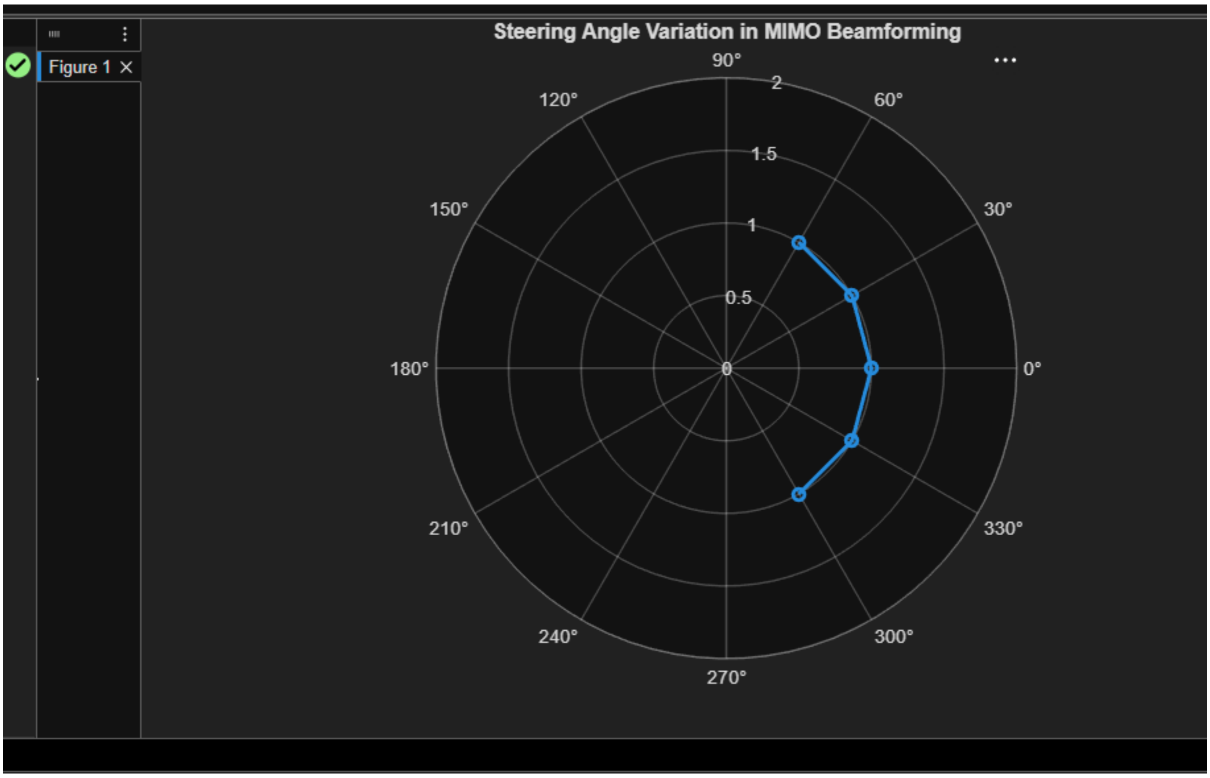
Ex No.8 - Steering Angle Variation in MIMO Beamforming Using MATLAB

Objective – 1 Analyze Steering Angle (Degrees)

MATLAB Code:

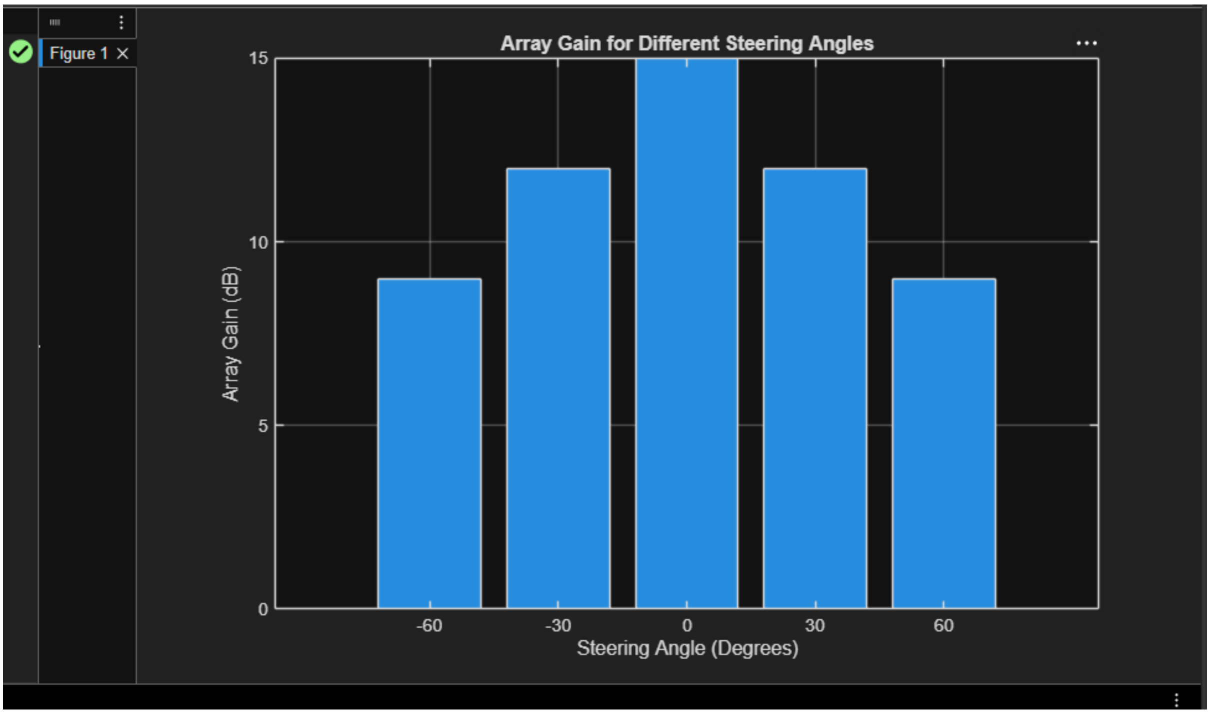
```
clc;

clear;

close all;

angle = [-60 -30 0 30 60];

gain = ones(size(angle));

figure

% Steering Angles (degrees)

% Normalized Gain

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```



Objective – 2 Compare Array Gain (dB)

MATLAB Code:

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60];

gain = [9 12 15 12 9];

figure

bar(angle, gain)

grid on

% Steering Angles (degrees)

% Array Gain (dB)

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')
```



Objective – 3 Evaluate Beam Direction Error (Degrees)

MATLAB Code:

```
clc;

clear;

close all;

desired = [-60 -30 0 30 60];

actual = [-58 -32 1 29 62];

error = abs(desired - actual);

figure

% Desired Steering Angle (degrees)

% Actual Beam Direction (degrees)

% Beam Direction Error

plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')
```